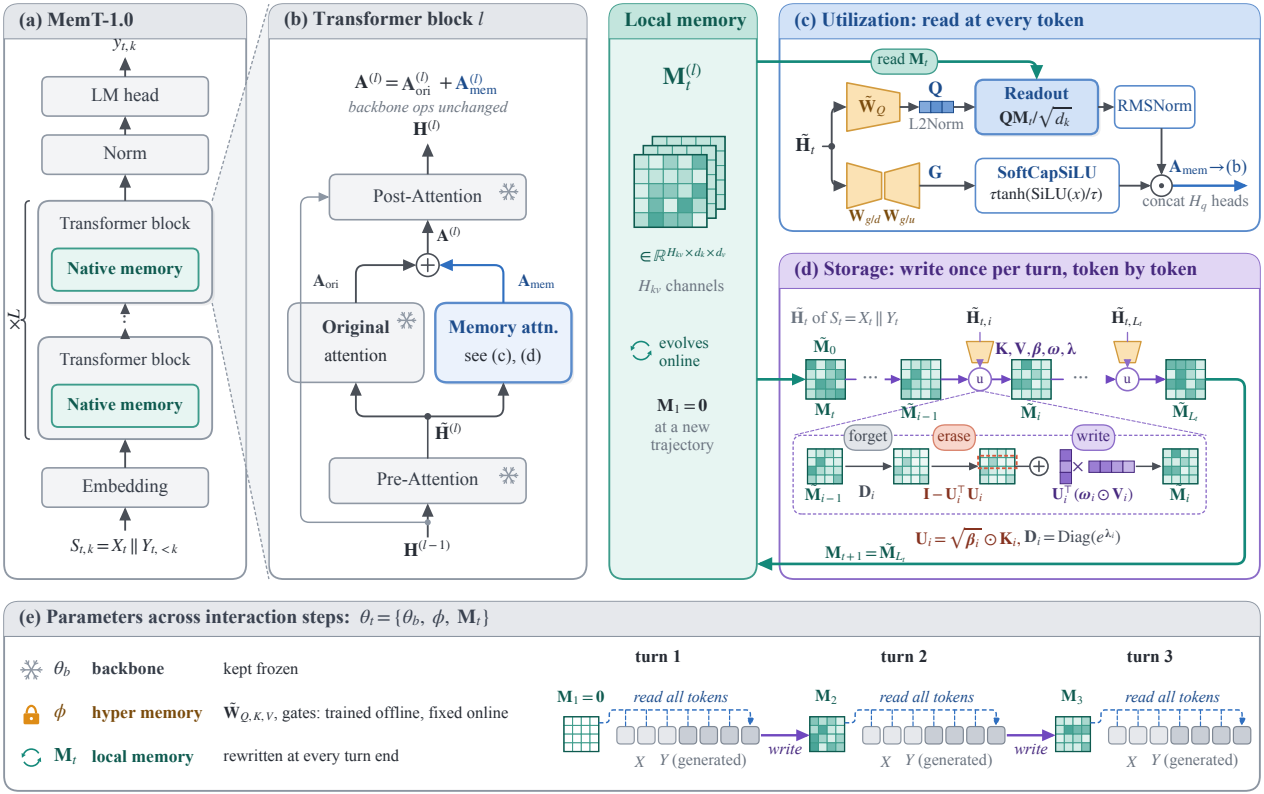

(a) MemT-1.0
$y_{t,k}$
LM head
Norm
Transformer block
Native memory
Transformer block
Native memory
Embedding
$S_{t,k} = X_t \,\|\, Y_{t,<k}$
×L
(b) Transformer block l
$\mathbf{A}^{(l)} = \mathbf{A}^{(l)}_{\text{ori}} + \mathbf{A}^{(l)}_{\text{mem}}$
backbone ops unchanged
$\mathbf{H}^{(l)}$
Post-Attention
$\mathbf{A}^{(l)}$
$\mathbf{A}_{\text{ori}}$
$\mathbf{A}_{\text{mem}}$
Original attention
Memory attn. see (c), (d)
$\tilde{\mathbf{H}}^{(l)}$
Pre-Attention
$\mathbf{H}^{(l-1)}$
Local memory
$\mathbf{M}_t^{(l)}$
$\in \mathbb{R}^{H_{kv} \times d_k \times d_v}$
H_{kv} channels
evolves online
$\mathbf{M}_1 = \mathbf{0}$ at a new trajectory
(c) Utilization: read at every token
read $\mathbf{M}_t$
$\tilde{\mathbf{H}}_t$
$\tilde{\mathbf{W}}_Q$
Q
L2Norm
Readout $\mathbf{Q}\mathbf{M}_t/\sqrt{d_k}$
RMSNorm
$\mathbf{A}_{\text{mem}} \rightarrow$ (b)
concat H_q heads
$\mathbf{W}_{g/d}$ $\mathbf{W}_{g/u}$
G
SoftCapSiLU $\tau\tanh(\text{SiLU}(x)/\tau)$
(d) Storage: write once per turn, token by token
$\tilde{\mathbf{H}}_t$ of $S_t = X_t \,\|\, Y_t$
$\tilde{\mathbf{H}}_{t,i}$
$\tilde{\mathbf{H}}_{t,L_t}$
$\tilde{\mathbf{M}}_0$
$\mathbf{M}_t$
$\mathbf{K}, \mathbf{V}, \boldsymbol{\beta}, \boldsymbol{\omega}, \boldsymbol{\lambda}$
u
$\tilde{\mathbf{M}}_{i-1}$
$\tilde{\mathbf{M}}_i$
$\tilde{\mathbf{M}}_{L_t}$
forget
erase
write
$\mathbf{M}_{i-1}$
$\mathbf{D}_i$
$\mathbf{I} - \mathbf{U}_i^\top \mathbf{U}_i$
$\mathbf{U}_i^\top (\boldsymbol{\omega}_i \odot \mathbf{V}_i)$
$\mathbf{M}_i$
$\mathbf{U}_i = \sqrt{\boldsymbol{\beta}_i} \odot \mathbf{K}_i$, $\mathbf{D}_i = \text{Diag}(e^{\lambda_i})$
$\mathbf{M}_{t+1} = \tilde{\mathbf{M}}_{L_t}$
(e) Parameters across interaction steps: $\theta_t = \{\theta_b, \phi, \mathbf{M}_t\}$
θ_b backbone kept frozen
ϕ hyper memory $\tilde{\mathbf{W}}_{Q,K,V}$, gates: trained offline, fixed online
$\mathbf{M}_t$ local memory rewritten at every turn end
turn 1
turn 2
turn 3
$\mathbf{M}_1 = \mathbf{0}$
read all tokens
X
Y (generated)
write
$\mathbf{M}_2$
$\mathbf{M}_3$